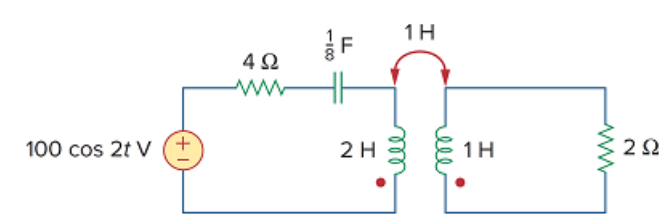


Problem

For the circuit in Fig. 13.18, determine the coupling coefficient and the energy stored in the coupled inductors at $t = 1.5$ s.

Figure 13.18



Step-by-step solution

Step 1 of 8

Refer to circuit diagram in Figure 13.18 in the textbook.

Calculate the coupling coefficient.

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

Substitute 1 H for M , 2 H for L_1 and 1 H for L_2 in the equation.

$$\begin{aligned} k &= \frac{1 \text{ H}}{\sqrt{(2 \text{ H})(1 \text{ H})}} \\ &= \frac{1}{\sqrt{2}} \\ &= \frac{1}{1.414} \\ &= 0.7071 \end{aligned}$$

Therefore, the value of coupling coefficient, k is 0.7071.

Step 2 of 8

The angular frequency is,

$$\omega = 2 \text{ rad/sec}$$

Calculate the impedance of L_1 .

$$\begin{aligned} X_{L1} &= j\omega L_1 \\ &= j(2)(2) \\ &= j4 \Omega \end{aligned}$$

Calculate the impedance of L_2 .

$$\begin{aligned} X_{L2} &= j\omega L_2 \\ &= j(2)(1) \\ &= j2 \Omega \end{aligned}$$

Step 3 of 8

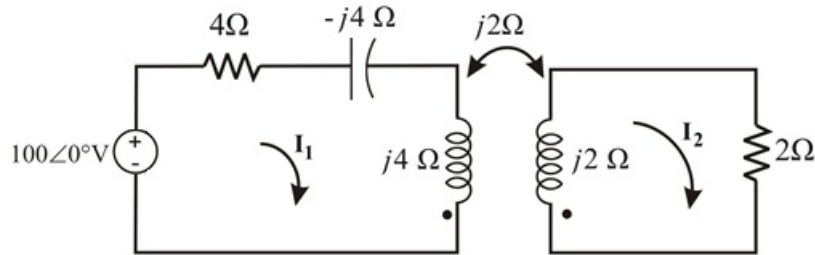
Calculate the impedance of M .

$$\begin{aligned} X_M &= j\omega L_2 \\ &= j(2)(1) \\ &= j2 \Omega \end{aligned}$$

Calculate the impedance of C .

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2)\left(\frac{1}{8}\right)} \\ &= -j4 \Omega \end{aligned}$$

Draw the frequency domain equivalent circuit with mesh notations.



Step 4 of 8

Apply Kirchhoff's voltage law to loop 1.

$$\begin{aligned} -100\angle 0^\circ + (4 - j4 + j4)\mathbf{I}_1 - j2\mathbf{I}_2 &= 0 \\ 4\mathbf{I}_1 - j2\mathbf{I}_2 &= 100\angle 0^\circ \dots\dots (1) \end{aligned}$$

Apply Kirchhoff's voltage law to loop 2.

$$\begin{aligned} 2\mathbf{I}_2 + j2\mathbf{I}_2 - j2\mathbf{I}_1 &= 0 \\ \mathbf{I}_2(2 + j2) &= j2\mathbf{I}_1 \\ \mathbf{I}_1 &= \left(\frac{2 + j2}{j2}\right)\mathbf{I}_2 \\ &= (1 - j1)\mathbf{I}_2 \dots\dots (2) \end{aligned}$$

Step 5 of 8

Recall equation (1).

$$4\mathbf{I}_1 - j2\mathbf{I}_2 = 100\angle 0^\circ$$

Substitute $(1 - j1)\mathbf{I}_2$ for $\mathbf{I}_1$ in the equation.

$$\begin{aligned} 4(1 - j1)\mathbf{I}_2 - j2\mathbf{I}_2 &= 100\angle 0^\circ \\ (4 - j4 - j2)\mathbf{I}_2 &= 100\angle 0^\circ \end{aligned}$$

$$\begin{aligned} \mathbf{I}_2 &= \frac{100\angle 0^\circ}{4 - j6} \\ &= \frac{100\angle 0^\circ}{7.21\angle -56.31^\circ} \\ &= 13.86\angle 56.31^\circ \text{ A} \end{aligned}$$

Step 6 of 8

Calculate the current $\mathbf{I}_1$.

$$\mathbf{I}_1 = (1 - j1)\mathbf{I}_2$$

Substitute $13.86\angle 56.31^\circ \text{ A}$ for $\mathbf{I}_2$ in the equation.

$$\begin{aligned}\mathbf{I}_1 &= (1 - j1)(13.86\angle 56.31^\circ \text{ A}) \\ &= (1.414\angle -45^\circ)(13.86\angle 56.31^\circ \text{ A}) \\ &= 19.61\angle 11.31^\circ \text{ A}\end{aligned}$$

The currents in time domain are,

$$i_1 = 19.61 \cos(2t + 11.31^\circ) \text{ A}$$

$$i_2 = 13.86 \cos(2t + 56.31^\circ) \text{ A}$$

Step 7 of 8

Calculate the current i_1 at $t = 1.5 \text{ s}$.

$$\begin{aligned}i_1 &= 19.61 \cos\left((2)(1.5 \text{ rad})\left(\frac{180^\circ}{\pi}\right) + 11.31^\circ\right) \text{ A} \\ &= 19.61 \cos(171.88^\circ + 11.31^\circ) \text{ A} \\ &= (19.61)(-0.9984) \text{ A} \\ &= -19.58 \text{ A}\end{aligned}$$

Calculate the current i_2 at $t = 1.5 \text{ s}$.

$$\begin{aligned}i_2 &= 13.86 \cos\left((2)(1.5 \text{ rad})\left(\frac{180^\circ}{\pi}\right) + 56.31^\circ\right) \text{ A} \\ &= 13.86 \cos(171.88^\circ + 56.31^\circ) \text{ A} \\ &= (13.86)(-0.666) \text{ A} \\ &= -9.25 \text{ A}\end{aligned}$$

Step 8 of 8

Calculate the total energy stored in the coupled inductors.

$$\begin{aligned}w &= \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 - M i_1 i_2 \\ &= \frac{1}{2}(2)(-19.58)^2 + \frac{1}{2}(1)(-9.23)^2 - (1)(-19.58)(-9.23) \\ &= \frac{1}{2}(2)(383.4) + \frac{1}{2}(1)(85.19) - (1)(180.72) \\ &= 383.4 + 42.6 - 180.72 \\ &= 426 - 180.72 \\ &= 245.3 \text{ J}\end{aligned}$$

Therefore, the total energy, w stored in the coupled inductors is 245.3 J